

KRASNOYARSK INTL, Russian Federation

WMO: 284935

Lat: **56.1795N**Lon: **92.4715E**Elev: **942**StdP: **14.20**Time Zone: **7.00 (E07)**Period: **99-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-37.8	-32.4	-43.3	0.5	-37.4	-38.0	0.6	-32.1	22.7	15.2	20.5	14.2	0.5	110	0.653

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.0	85.7	66.5	82.3	64.7	78.8	63.0	68.9	80.4	67.1	78.0	65.1	75.7	7.5	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.7	94.8	73.1	62.8	88.6	71.6	61.0	82.9	69.9	33.5	79.5	31.9	77.9	30.6	75.5	81.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.8	18.2	16.1	-38.1	91.4	8.1	3.1	-44.0	93.6	-48.7	95.4	-53.3	97.2	-59.2	99.4		
(5)			-38.3	72.1	8.1	3.0	-44.1	74.2	-48.8	75.9	-53.3	77.6	-59.2	79.7	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	33.6	-0.6	5.4	22.1	36.3	48.6	61.5	64.6	60.0	48.2	34.2	16.8	4.9		
	DBStd	25.32	17.34	15.35	11.39	9.32	9.51	7.02	5.17	6.45	8.00	10.13	14.61	16.56		
	HDD50	7278	1568	1249	864	419	146	7	0	5	129	496	997	1398		
	HDD65	11574	2033	1669	1329	861	515	144	70	181	506	956	1447	1863		
	CDD50	1309	0	0	0	7	101	353	454	314	74	6	0	0		
	CDD65	128	0	0	0	0	5	39	58	25	1	0	0	0		
	CDH74	2010	0	0	0	2	154	696	743	368	47	0	0	0		
	CDH80	514	0	0	0	0	37	180	195	94	8	0	0	0		
	WSAvg	6.2	5.0	5.9	7.5	8.2	7.8	5.5	4.7	4.7	5.9	6.8	6.6	5.7		
	PrecAvg	19.9	0.9	0.8	0.8	1.1	1.9	2.3	2.1	3.0	2.1	1.8	1.6	1.3		
	PrecMax	24.9	1.8	1.5	1.9	2.2	3.7	3.8	4.5	5.5	4.0	3.3	2.4	2.4		
(15) Precipitation	PrecMin	15.2	0.0	0.2	0.0	0.3	0.5	0.7	0.3	1.3	0.6	0.6	1.0	0.6		
	PrecStd	2.8	0.5	0.4	0.5	0.6	1.1	0.9	1.0	1.1	0.9	0.7	0.4	0.6		
	0.4% DB	35.4	39.1	51.9	70.0	84.6	89.3	89.7	87.7	80.4	64.3	51.5	37.5			
	0.4% MCWB	30.9	33.2	39.9	50.0	64.0	67.1	68.4	68.8	62.4	50.3	43.4	31.5			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	30.3	34.2	46.2	62.7	78.5	84.5	85.6	82.4	73.0	58.9	44.4	33.4			
	2% MCWB	26.7	28.9	37.2	47.5	58.7	65.2	67.0	66.5	58.9	48.4	38.5	29.4			
	5% DB	26.9	31.6	42.4	57.6	73.3	81.0	82.0	78.4	67.7	53.6	39.4	30.1			
	5% MCWB	23.8	27.5	35.1	45.1	55.6	63.5	65.5	64.2	56.1	45.6	35.0	27.1			
	10% DB	24.1	27.3	37.8	53.3	68.1	78.4	78.4	73.6	62.5	48.6	35.6	26.8			
	10% MCWB	21.5	24.1	32.5	42.7	52.7	61.9	64.0	61.9	53.3	43.2	32.5	23.9			
	0.4% WB	31.0	33.3	41.8	52.3	65.6	70.2	72.5	71.1	63.6	53.1	43.9	32.6			
	0.4% MCDB	33.4	37.4	49.6	65.3	81.8	83.0	83.4	83.8	78.1	61.2	49.7	35.9			
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	26.8	29.9	38.0	49.3	60.5	67.6	69.3	67.8	59.8	49.4	38.7	29.5			
	2% MCDB	29.8	33.8	44.6	60.7	73.8	80.2	79.9	78.1	70.4	57.3	43.6	32.8			
	5% WB	24.0	27.1	35.3	46.3	57.2	65.6	67.4	65.2	57.3	46.6	35.0	26.9			
	5% MCDB	27.0	30.8	41.1	56.6	69.4	78.0	77.4	75.6	65.7	52.5	38.7	29.7			
	10% WB	21.1	24.1	32.7	42.8	54.2	63.6	65.6	63.1	54.6	43.4	32.4	24.0			
	10% MCDB	23.8	27.5	37.6	52.0	66.3	74.7	75.0	71.9	61.4	48.5	35.4	26.9			
	MDBR	16.9	18.8	19.9	20.4	23.7	24.5	22.0	21.7	19.6	15.8	15.0	15.8			
	5% DB	17.0	19.4	23.6	30.1	34.0	31.2	29.1	29.8	29.9	23.8	15.9	15.7			
(35) Mean Daily Temperature Range	MCWBR	15.0	16.1	16.6	17.6	18.2	14.2	13.4	15.3	16.9	15.0	11.6	13.7			
	MCDBR	16.7	18.3	21.9	28.2	29.6	26.8	24.3	26.6	26.0	22.0	15.1	15.8			
	MCWBR	14.9	15.6	16.1	17.6	17.3	13.6	12.9	14.4	15.5	14.5	11.4	14.2			
	MCDBR	14.9	15.6	16.1	17.6	17.3	13.6	12.9	14.4	15.5	14.5	11.4	14.2			
(40) Clear-Sky Solar Irradiance	taub	0.283	0.316	0.362	0.392	0.411	0.407	0.430	0.410	0.369	0.350	0.317	0.283			
	taud	2.020	2.002	1.957	2.120	2.200	2.305	2.253	2.276	2.342	2.236	2.057	1.964			
	Ebn at Noon	182	225	244	259	262	265	255	252	247	215	165	144			
	Edh at Noon	21	32	43	43	42	38	40	36	30	25	20	17			
(44) All-Sky Solar Radiation	RadAvg	248	561	990	1365	1562	1885	1685	1347	861	453	224	146			
	RadStd	28	37	73	89	164	157	112	117	114	35	30	19			

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (20 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page